

Sizing and simulation of a photovoltaic installation at the Al-Hoceima desalination plant

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ARTICLE INFO

Keywords:

Desalination plant
Photovoltaic system
PVsyst software
Sizing
Simulation
Energy consumption

ABSTRACT

One of the major challenges in Morocco is the high energy consumption of desalination plants, coupled with the growing concern over climate change. This has driven the country to explore more sustainable solutions, such as the use of renewable energy, particularly solar power. In this context, we designed a photovoltaic system using PVsyst software to supply the Al-Hoceima seawater desalination plant, with a specific focus on the reverse osmosis unit, as it is the most energy-intensive part of the facility. This paper presents the total energy consumption of the Al-Hoceima desalination plant, the sizing of the PV installation, the peak power capacity used, and a simulation of normalized production and performance ratio, along with the system's energy losses. Finally, an economic and environmental analysis is conducted, detailing the payback period (5,59 years) and the greenhouse gas emissions (GHG) avoided by the system (1830,25 ton/year). The results of this study show that, after installing the photovoltaic system, we achieved an approximate reduction of 10 % in the total energy consumption of the Al-Hoceima desalination plant.

1. Introduction

The rapid increase in water requirements resulting from population growth is driving the exploitation of non-conventional water resources, including the desalination of seawater [1]. In light of the mounting global demand for water, the primary objective of all operators is to produce potable water of satisfactory quality at the lowest possible cost. One of the primary disadvantages of desalination as a source of drinking water is its high energy consumption [2]. At present, the energy expenditure associated with the desalination of seawater by reverse osmosis can account for up to 50 % of the total production cost [3]. Reducing specific energy consumption represents a significant challenge in meeting the growing demand for improved quality and lower costs. In Morocco, environmental and energy issues are becoming increasingly prominent concerns. As the population grows and living standards improve, energy consumption increases accordingly. This has prompted Morocco to pursue measures to enhance energy security in recent years. In order to pursue development and modernization and to combat vulnerability, it is essential that Morocco achieve security and

independence of energy supply. In this regard, the Moroccan government has set a national target of producing 42 % of its energy from renewable resources such as hydropower, solar and wind energy, as well as reducing GHG (Greenhouses Gas) emissions by 2874 KT (Kiloton) of CO₂ per year by 2030 [4]. This is a challenging objective for a nation in the process of development. It is plausible that Morocco has identified an additional crucial element for energy conservation, namely the strategy of enhancing energy efficiency, particularly at the level of buildings and companies [5]. This can be achieved through the implementation of duly approved policies and investments aimed at reducing energy costs. It is therefore crucial to reduce energy dependency by adopting energy efficiency as an alternative source of energy in a range of establishments. In order to enhance the quality and security of drinking water in the town of Al-Hoceima and its surrounding area, ONEE (National Office of Electricity and Drinking Water) has established a desalination facility with a production capacity of 200 liters per second (l/s). The reverse osmosis technique is employed at this station to extract salt and impurities from seawater [6]. While an efficient process, it nevertheless requires a significant amount of energy. Nevertheless, this process is

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<https://doi.org/10.1016/j.dwt.2025.101171>

Received 2 February 2025; Received in revised form 6 April 2025; Accepted 9 April 2025

Available online 10 April 2025

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highly energy-intensive, thereby giving rise to concerns pertaining to the energy efficiency and environmental impact associated with the utilization of fossil fuels to power these infrastructures [7]. In this context, the integration of renewable energies, especially solar energy, presents an opportunity to enhance the energy efficiency of desalination plants while reducing their ecological footprint [8]. The Al-Hoceima region is characterized by a high degree of sunshine, which renders it conducive to the implementation of solar energy systems [9]. The primary objective of this study is to devise a solar energy system that can serve as a source of power for the desalination plant in Al-Hoceima. Through this investigation, we seek to evaluate the plant's existing energy efficiency and propose a renewable solution that is tailored to its energy requirements, while also contributing to the transition towards a low-carbon economy. Recent studies have emphasized the potential of integrating renewable energy into desalination technologies to address challenges related to high energy consumption and environmental impact. For instance, one study examined the utilization of hydrophilic candle waste microcapsules as a thermal energy storage medium, with the objective of enhancing the cogeneration of steam and electricity. This approach enables continuous operation while concurrently reducing energy consumption [10]. Concurrently, an additional investigation conducted a thermo-economic analysis of a reverse osmosis desalination unit powered by photovoltaic/thermal (PVT) panels and preheated using geothermal energy, demonstrating a substantial reduction in specific energy consumption and an enhancement in water production [11]. In a further study, researchers proposed a small-scale PVT-RO system with integrated cooling and preheating mechanisms for use in isolated regions, thereby showcasing its potential to sustainably provide both potable water and electricity [12]. These innovative approaches underscore the feasibility and strategic interest of renewable-powered desalination, offering promising avenues to enhance the energy efficiency and sustainability of desalination infrastructures in Morocco. The novelty of our work lies in its integrated approach, which combines detailed photovoltaic system sizing using PVsyst with an in-depth analysis of both economic and environmental impacts. This approach is specifically tailored to the Al-Hoceima desalination plant. By leveraging real energy consumption data alongside the region's abundant solar resources, this study provides a pragmatic, context-specific evaluation of renewable energy integration in desalination. This innovative methodology not only addresses an existing gap in the literature but also offers a framework that can be applied to other developing regions facing similar challenges in energy security and environmental sustainability.

2. State of the art

2.1. Energy consumption estimate for Al-Hoceima desalination plant

2.1.1. Electricity consumption in kWh

The production of an electrical balance sheet for an industrial unit represents the initial phase of a process aimed at reducing electricity consumption [13]. It serves as an indispensable tool for identifying the components of the Al-Hoceima desalination plant that consume the greatest amount of energy. Table 1 presents the estimated energy consumption in kWh/day and the associated cost for the Al-Hoceima desalination plant in DH (Dirham). The energy contribution of each unit was determined by measuring the power absorption of the equipment multiplied by the operational time.

$$E = P \times \Delta t$$

Where:

E = Electrical energy consumed (Joule or watt-hour)

P = Power of the equipment (Watt)

Δt = Operation duration (second or hour)

The consumption cost was calculated according to three periods of operation: peak hours, off-peak hours, and full hours.

Table 1

Estimated Electricity Consumption of the Desalination Plant in Al-Hoceima.

	Total Active Consumption (kWh)	Consumption Cost (DH)
Pumping on boreholes	4800	5134,8
Raw water pumping	3780	4043,66
Pretreatment	34,95	34,32
Reverse Osmosis	24873,60	26608,53
CIP	0,00	0,00
Treated water Pumping	3180	3401,81
Neutralisation	15,95	17,06
Brine pumping	1555,20	1663,68
CO2	9,48	10,14
Reagent building	8,88	9,5
Chlorine	45	47,22
Water for diluting reagents	60	62,96
Service air	150,71	161,21
Preparation of lime milk	35	36,73
Lime saturator	51	54,25
Administration	966,82	933,39
Industrial air conditioning	7581,2	7366,28
Total	47147,79	49585,53

The energy consumption of the desalination plant is approximately 47147,79 kWh/day, resulting in a total cost of 49585,53 DH. To facilitate a more comprehensive understanding of the contribution of each unit to the station's overall energy consumption, the percentage of each unit has been presented in Fig. 1.

The majority of electricity is consumed by the reverse osmosis unit (53 %), followed by industrial air conditioning (16 %), borehole pumping (10 %), raw water pumping (8 %) and treated water pumping (7 %). However, the proportion of electricity consumption attributed to brine pumping and administrative equipment is relatively low, at 3 % and 2 % respectively. This distribution is based on the power and number of operating hours of the desalination plant. The reverse osmosis unit requires a total operating power of approximately 2072,8 kW in order to provide a pressure of 57 bars to the membrane feed water flow. This unit consumes a total of 24873,60 kWh/day, including 22800 kWh/day for the high-pressure (HP) pump and 2073,60 kWh/day for the booster pump. This represents over half of the total energy consumption of the desalination plant. The deployment of industrial air conditioning at the station serves the dual purpose of safeguarding production equipment from the risks of overheating and explosions. While the aggregate power consumption of industrial air conditioning is relatively modest in comparison to that of the reverse osmosis unit, the continuous operation of the former (24 hours per day) results in considerable energy expenditure. The percentages indicated for the units of pumping on drilling, raw water pumping, and treated water pumping are approximate due to the fact that the total power in service is also approximately 400 kW, 315 kW, and 265 kW, respectively.

2.1.2. Electricity consumption in CO2 equivalent

The carbon dioxide (CO₂) content of the electrical kilowatt-hour represents the quantity of CO₂ emissions generated by the production of one kilowatt-hour of electricity [14]. The quantity of electricity consumed at the Al-Hoceima desalination plant, expressed in CO₂ equivalents, can be calculated using the following formula:

$$GHGE = \frac{\text{kgCO}_2\text{equivalent}}{\text{Area in m}^2}$$

The production of 1 kWh in Morocco results in CO₂ emissions with a value of 0.728 kg CO₂ eq. Despite the desalination plant covering an area of approximately 3.2 ha and consuming 17208944,66 kWh/year, the rate of greenhouse gas emissions is 398.33 kg CO₂ eq/year/m² Fig. 2.

The objective of this label is to enhance awareness and provide information about greenhouse gases (GHG) in order to facilitate their consideration in heating modes and the selection of energy sources for

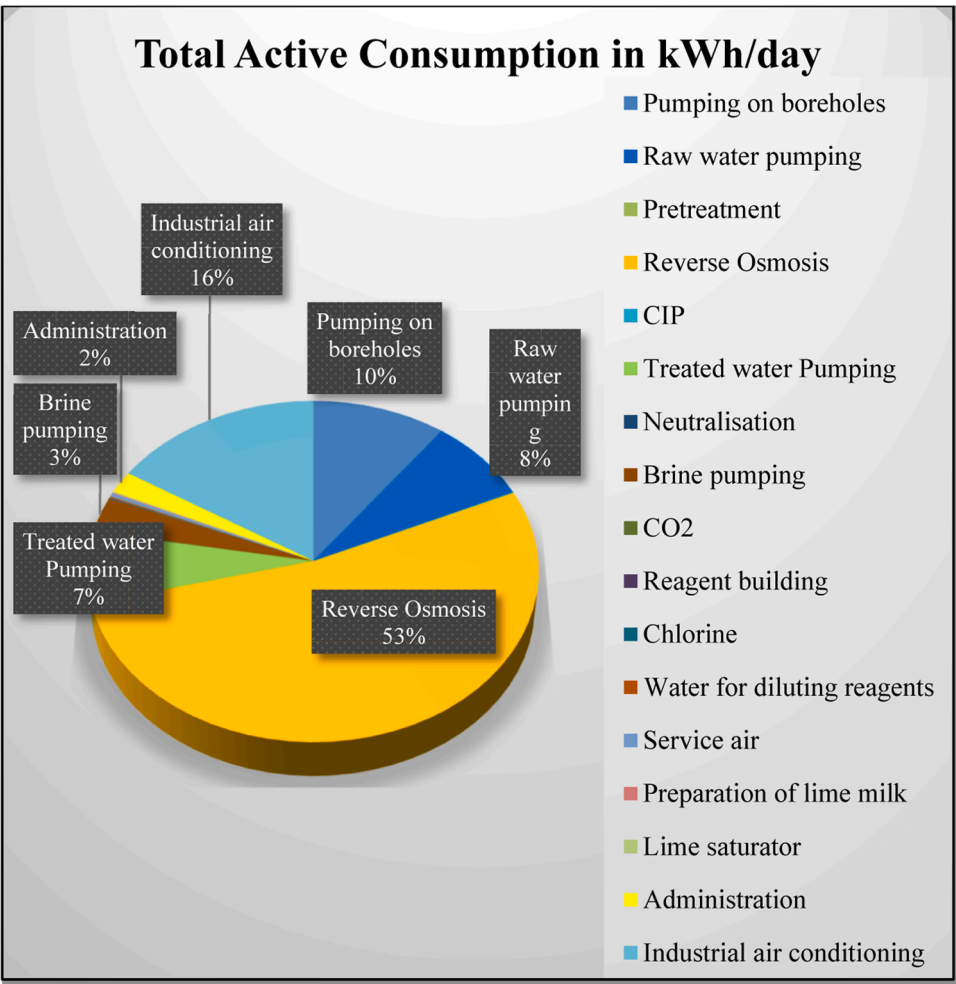


Fig. 1. Daily Energy Consumption of the Different Units of the Desalination Dlant in Al-Hoceima.

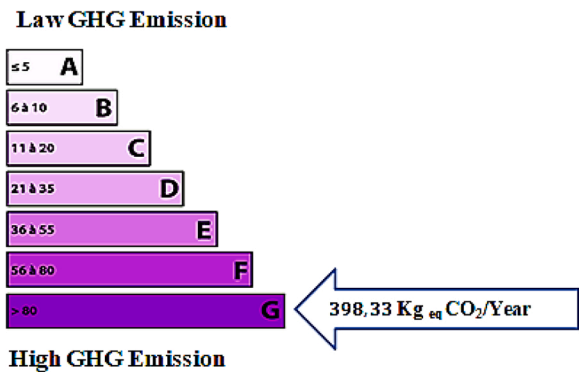


Fig. 2. Greenhouse Gas Emission Rate of Al-Hoceima Desalination Plant.

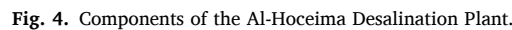
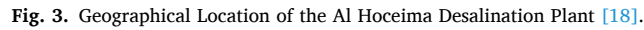
housing, as well as in any related work [15]. In accordance with the specifications outlined on the energy label, the level of greenhouse gas emissions for the desalination plant is classified as level G. This classification necessitates the implementation of an energy management system. The implementation of a renewable energy project represents a potential solution to the reduction of greenhouse gases and electricity costs. When considering energy-saving projects that utilize renewable energies, solar energy is often the first option that comes to mind. This renewable source has the potential to generate 1000 times more energy than other renewable sources combined [16], and is several orders of magnitude higher than the current total global electricity demand.

3. Materials and methods

3.1. Presentation of the study area

The Al-Hoceima seawater desalination plant plays a crucial role in northern Morocco, situated along the Mediterranean coast within the Tanger-Tetouan-Al Hoceima region [17]. Located 8 km southeast of Al-Hoceima, the facility is positioned near Sfiha Beach in the commune of Ajdir, covering a total area of 3.2 ha on land expropriated for the benefit of ONEE’s water branch (Fig. 3).

The desalination plant in Al-Hoceima utilizes reverse osmosis as a membrane technology for converting seawater into potable water. With a production capacity of 200 l/s, or 17,280 m³/day, the facility plays a crucial role in providing a consistent supply of drinking water to satisfy the region’s demands. This output is essential for meeting the vital needs of the local population and supporting economic activities. The plant features various water treatment systems, including sand filters, cartridge filters, and reverse osmosis membranes. Prior to entering the reverse osmosis membranes, seawater undergoes pretreatment, which includes a filtration process to eliminate ions and dissolved salts. The treated water is then subjected to additional processing to remove both organic and inorganic pollutants before it is stored in tanks for distribution to consumers. Fig. 4 illustrates the components of the Al-Hoceima desalination plant.



P_P: Peak power (kWp);
E_C: Daily energy consumption (kWh/day);
K: Conversion factor (the coefficient k varies from 0.85 for well-

The setup of a photovoltaic system is influenced by the site and the level of solar irradiation it receives [20]. The performance, alignment, and tilt of the solar panels typically depend on the type of modules and inverters used. Weather conditions such as solar radiation, wind speed, and ambient temperature are monitored to assess the site's suitability for the system [21]. The PVsyst software is utilized in this analysis for

simulation purposes. PVsyst is a comprehensive software application designed for the analysis, design, and evaluation of complete photovoltaic (PV) systems [22]. It accommodates a variety of system types, including grid-connected, off-grid, pumping, and direct current (DC) networks, such as those used in public transportation. The software features comprehensive databases for both meteorological conditions and PV components, as well as general tools for solar energy assessment. Tailored for architects, engineers, and researchers, PVsyst also serves as an effective educational resource [23]. The software provides users with robust preliminary design tools, allowing for rapid evaluations of system performance based on a few general parameters, without needing to specify actual components. Users can also obtain rough cost estimates. For off-grid systems, PVsyst helps determine the necessary PV power and battery capacity according to user load profiles and acceptable risk levels for power shortfalls. In the case of water pumping systems, it evaluates the required PV power based on water demand and pumping depth, considering the probability of unmet needs throughout the year [24]. For detailed project design and simulation, PVsyst allows for extensive hourly simulations, enabling users to define specific system components, assess shading impacts, and analyze performance metrics like energy yield, losses, and economic feasibility. The software supports various meteorological and component databases, providing tools for data visualization and optimization, making PVsyst a vital resource for accurate and professional PV system design and analysis [21].

4. Results and discussion

4.1. Project needs assessment

As part of an energy efficiency initiative at the desalination plant, a grid-connected photovoltaic system was designed using PVsyst software. This tool allows for the optimal configuration of the system by specifying key parameters such as the installed power capacity, available surface area, type of photovoltaic panels, and inverters. The primary objective is to reduce a portion of the energy consumption associated with the reverse osmosis unit, which is identified as the most energy demanding component, accounting for over 52 % of the station's total electrical consumption. By considering the available surface area, the aim is to reduce the unit's current consumption of 4651 kWh/day, representing approximately 18 % of the total energy use of the reverse osmosis system (which presents 10 % of total energy consumption of the plant). So, we can calculate the peak power [25], and in our case we found:

$$P_p = \frac{E_c}{K \times I_r} = \frac{4651}{0.75 \times 4.83} = 1284 \text{ kW}_p$$

where E_c represents the daily energy to be covered by the PV system (4651 kWh), K is the performance ratio of the PV system (assumed to be 0.75), and I_r is the average daily solar irradiation in Al-Hoceima (4.83 kWh/m²/day). Based on the calculation provided in your study, the peak power capacity of the photovoltaic (PV) system required to supply part of the energy needs of the Al-Hoceima desalination plant is 1284 kW_p.

4.2. Dimensioning of a solar energy system by PVsyst

In this study, the sizing of the solar energy system for the desalination plant was carried out using the PVsyst software, which is widely used for the design and simulation of photovoltaic systems. The photovoltaic system was sized based on the energy consumption of the reverse osmosis unit, which accounts for more than 50 % of the plant's total consumption. The objective was to supply approximately 18 % of this unit's consumption, or 4651 kWh/day. To this end, a thorough analysis of the reverse osmosis unit's energy demand was conducted, taking into account its operational hours and seasonal variations. The system was designed with a peak power of 1284 kW_p, ensuring that the solar power generated would meet the energy demands of the reverse osmosis unit.

The PVsyst software was utilized to determine the optimal number of solar panels and their arrangement, with inputs including local solar radiation data, module specifications, and system losses. This comprehensive approach ensured a reliable, efficient, and sustainable energy solution. The system was designed to cover 18 % of the reverse osmosis unit's consumption throughout the year, ensuring that the photovoltaic installation could contribute significantly to the plant's energy needs. The simulations were executed using PVsyst software, incorporating local meteorological data such as solar irradiance, temperature, and seasonal variations. The software also accounted for the technical specifications of the photovoltaic components, including modules and inverters, to accurately model the system's performance. These simulations provide reliable theoretical projections regarding the system's energy production and efficiency. It is imperative to acknowledge that the present simulations have not been subjected to experimental validation.

4.2.1. Geographical and meteorological data of the project site

All points on the Earth's surface are identified by a set of geographic coordinates, including longitude, latitude, and altitude. The Al-Hoceima desalination station is situated at 35.2008° north, 3.8941° east longitude, as illustrated in Fig. 5.

The design of photovoltaic systems necessitates an understanding of the solar radiation that is beneficial at the installation site. The meteorological parameters, including the monthly values of global irradiation, diffused irradiation, temperature, and wind speed, are presented in Fig. 6. Consequently, the mean value of the iridescence is approximately 4,83 kWh/m³/day.

4.2.2. Orientation of the panels

The orientation and inclination of photovoltaic modules are key factors influencing their energy production. Proper positioning is essential to ensure maximum efficiency [26]. Orientation refers to the direction the active surface of the panel faces (South, North, South-west), while the tilt angle is the inclination of the panel relative to the horizontal plane, typically measured in degrees. For our system, we selected a fixed tilt of 32° (Fig. 7), determined as optimal by the PVsyst software for maximizing annual solar exposure. The panels are oriented due south, achieving near-optimal performance with minimal energy losses.

4.2.3. System sizing

At the desalination station, we have an available area of approximately 0.76 ha designated for the installation of photovoltaic panels (Fig. 8). This area is sufficient for the installation of photovoltaic panels to achieve a usable power output of approximately 1284 kW_p.

For this project, the following components have been selected:

- Panels: JKM410M-72-V
- Inverters: Sunny High-power SHP150-US-20-PEAK3

Table 2 and Table 3 illustrate the characteristics of the selected PV module and inverter used, respectively.

The screenshot displays the 'Geographical coordinates of site' form in PVsyst. It features two columns: 'Decimal' and 'Deg. Min. Sec.'. The 'Latitude' row shows '35.2008' in decimal and '35', '12', '2' in degrees, minutes, and seconds, with a note '(+ = North, - = South hemisph.)'. The 'Longitude' row shows '-3.8941' in decimal and '-3', '53', '38' in degrees, minutes, and seconds, with a note '(+ = East, - = West of Greenwich)'. The 'Altitude' row shows '0' in decimal and 'M above sea level'. The 'Time zone' row shows '0.0' in decimal and 'Corresponding to an average difference', with a note 'Legal Time - Solar Time = 0h 16m'. A 'Get from name' button is located at the bottom.

Fig. 5. Geographical coordinates of site.

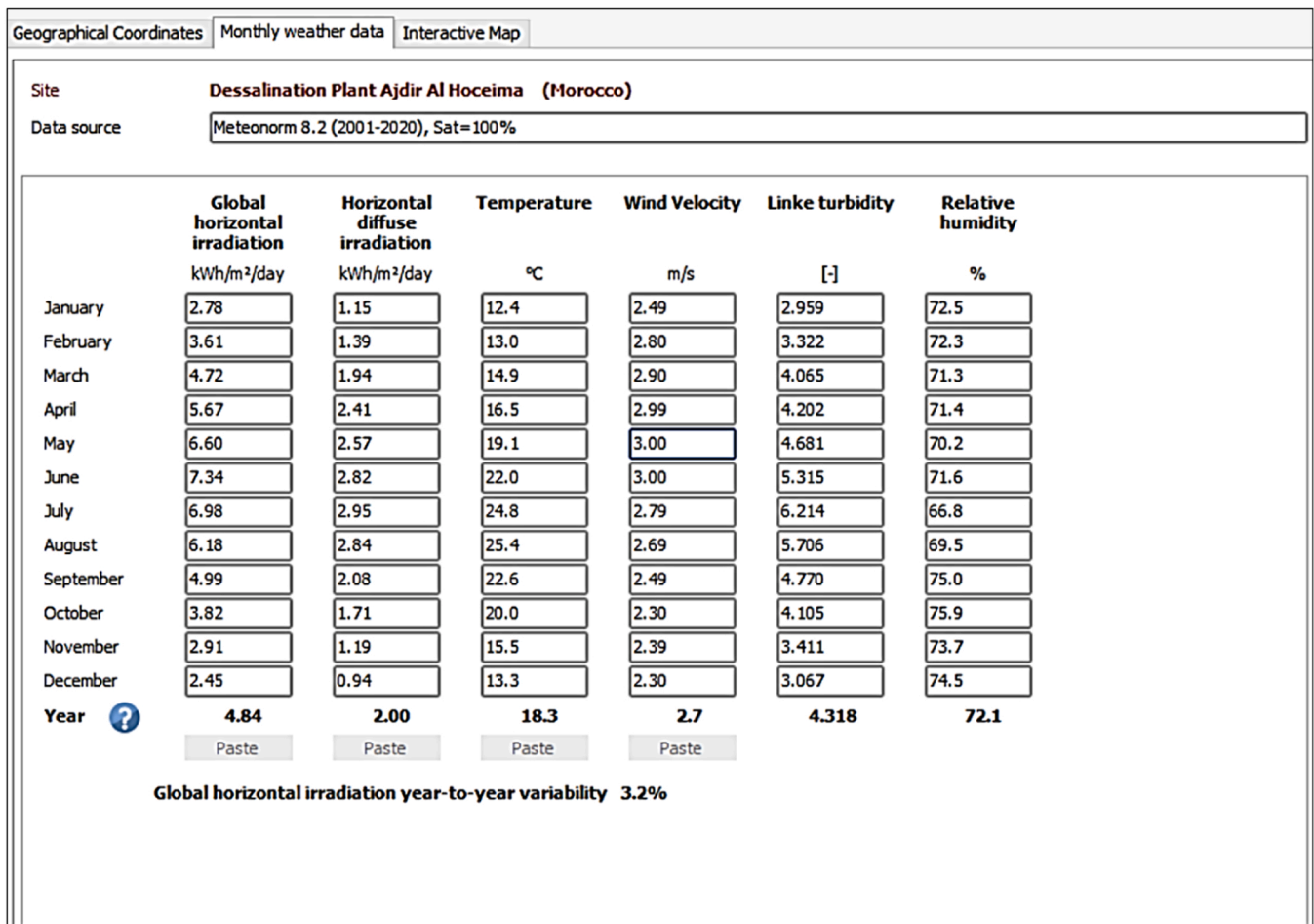


Fig. 6. Site weather data.

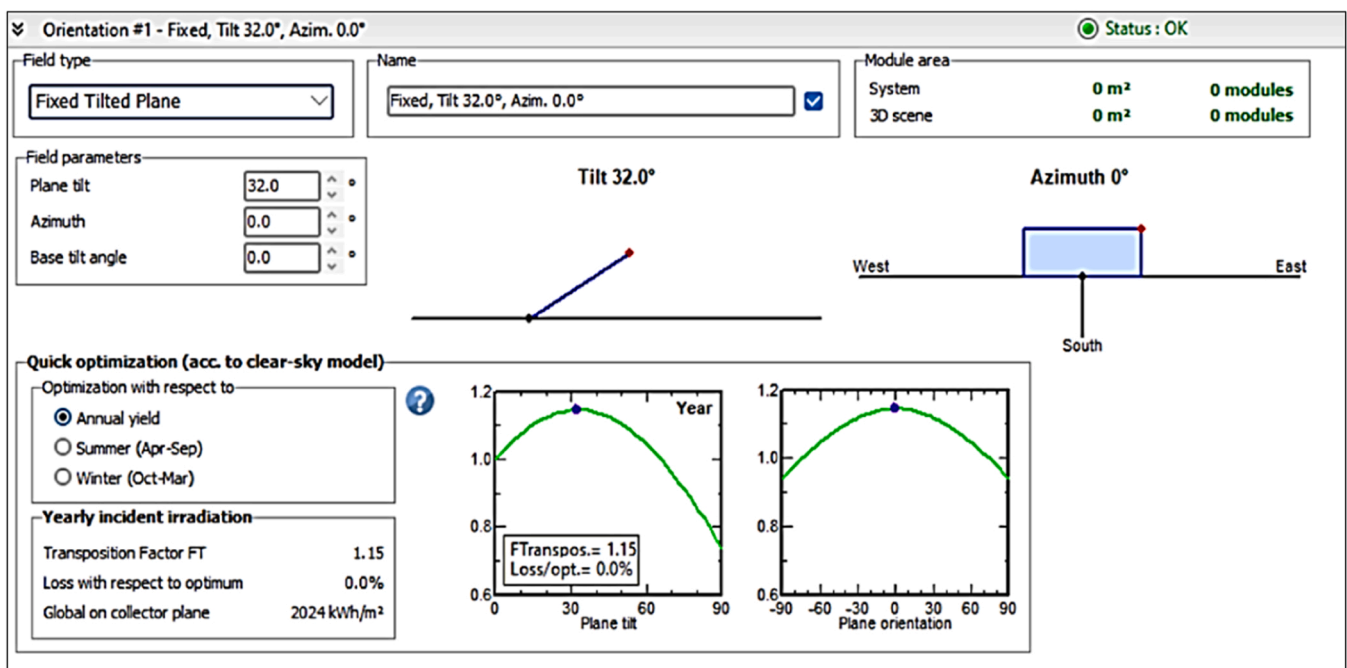


Fig. 7. Influence of Inclination and Orientation on Energy Collection.



Fig. 8. Available Area for Installation of Photovoltaic Panels.

Table 2
PV Module Characteristics.

Brand	Jinkosolar
Model	JKM410M-72H-V
Cell type	Monocrystalline
Peak power	410Wp
Efficiency	20.38 %
Material warranty	12 ans
Performance warranty	12 ans
Dimensions	2008 × 1002 × 40 mm
Unit price	74,55 €

Table 3
Technical Specifications of the Selected Inverter.

Brand	SMA
Model	Sunny Highpower SHP 150-US-20-PEAK3
Efficiency	98.8–98.6 %
U max	1500 V
I max	180 A
Dimensions	770 × 830 × 444 mm
Unit price	6799 €

The selection of panels and inverters was based on their availability in the local market and their compliance with the specified technical requirements. It should be noted that PVsyst will suggest a module/system configuration that allows for an initial preliminary simulation [27]. The software includes a color-coded error or warning system to notify users of any inconsistencies, errors, or warnings, which will be displayed in the corresponding window. As a result, the PVsyst software shows that we can use 3132 PV modules (116 in parallel x 27 in series) with a unit power of 410 W_p. The system will also require 8 inverters, each with a unit power of 150 kW (Fig. 9).

4.2.4. Path of the sun

A fundamental understanding of the apparent movement of the sun for a specific point on the Earth's surface is essential for the development of any solar-based application [28]. The position of the sun is defined by two angles: its height (the angle between the sun and the horizontal plane of the location) and its azimuth (the angle with the south direction, counted negative to the east). Fig. 10 illustrates the height of the sun in the sky as a function of the azimuth at Ajdir (location of desalination station) over the course of a year.

4.2.5. Energy production and performance ratio

After entering the site data, the simulation calculates the energy distribution over the course of the year. PVsyst software presents the main results, including the expected annual and monthly energy production (Fig. 11). According to these results, the annual energy production is approximately 2256 MWh/year. The energy produced throughout the year reaches its peak during the summer months (June and July) and its lowest levels in the winter period (December). It is noted that a portion of this production is lost in connections and inverters, while additional losses occur in the photovoltaic panels due to the variation in the sun's elevation, which impacts the angle of incidence of solar irradiation. The performance ratio (PR), defined as the ratio of the system's output (Y_f) to the reference incident energy (Y_r) [24], is PR = 0.876, equivalent to 87.61 %. This PR value of 87.61 % is a strong indicator of the system's efficiency, as PR values above 80 % are generally considered good for photovoltaic systems. This suggests that the system is expected to operate with relatively low losses, which is crucial for ensuring a reliable energy supply for the desalination process. The system's performance is found to be close to its theoretical potential, with factors such as system losses, temperature variations, and module efficiency being taken into account.

The specific production value of 1757 kWh/kW_p/year reflects a solid performance in terms of energy yield per installed capacity, which aligns well with the expected energy requirements of the reverse osmosis unit.

List of subarrays

Name	#Mod #Inv.	#String #MPPT
PV Array		
Jinkosolar - JKM-410M-72H-V	27	116
SMA - Sunny Highpower SHP150-L	8	1

Sub-array **PV Array** **Add** **Delete**

Select the orientation
Orientation: Fixed, Tilt 32.0°, Azim. 0.0°

Pre-sizing Help
☐ No sizing
☒ Resize
 Planned power: 1284 kWp
 or available area: 6302 m²

Select the PV module
 Available Now: Filter: All PV modules
 Jinkosolar 410 Wp 36V Si-mono JKM-410M-72H-V Since 2020 Datasheets 2020 **Open**
 Approx. needed modules: 3132
☐ Use optimizer
 Sizing voltages: Vmpp (60°C) 35.8 V
 Voc (-10°C) 55.5 V

Select the inverter
 Available Now: Output voltage 600 V Tri 60Hz
 SMA 150 kW 855 - 1500 V TL 60 Hz Sunny Highpower SHP150-US-21-PEAK3 Since 2022 **Open**
 Nb. of inverters: 8 ☐
 Operating voltage: 855-1500 V Global Inverter's power: 1200 kWac
 Input maximum voltage: 1500 V

Design the array
Number of modules and strings
 Mod. in series: 27 ☐ between 24 and 27
 Nb. strings: 116 ☒ between 108 and 153
 Overload loss: 0.0 %
 Pnom ratio: 1.07 **Sizing**
 Nb. modules: 3132 Area: 6302 m²

Operating conditions
 Vmpp (60°C) 966 V
 Vmpp (20°C) 1135 V
 Voc (-10°C) 1499 V
 Plane irradiance: 1000 W/m²
 Impp (STC) 1124 A
 Isc (STC) 1230 A
 Isc (at STC) 1230 A
☐ Max. in data ☒ STC
 Max. operating power (at 1000 W/m² and 50°C): 1175 kW
 Array nom. Power (STC): 1284 kWp

Global system summary

Nb. of modules	3132
Module area	6302 m ²
Nb. of inverters	8
Nominal PV Power	1284 kWp
Nominal AC Power	1200 kWac
Pnom ratio	1.070

Fig. 9. System Configuration in PVsyst.

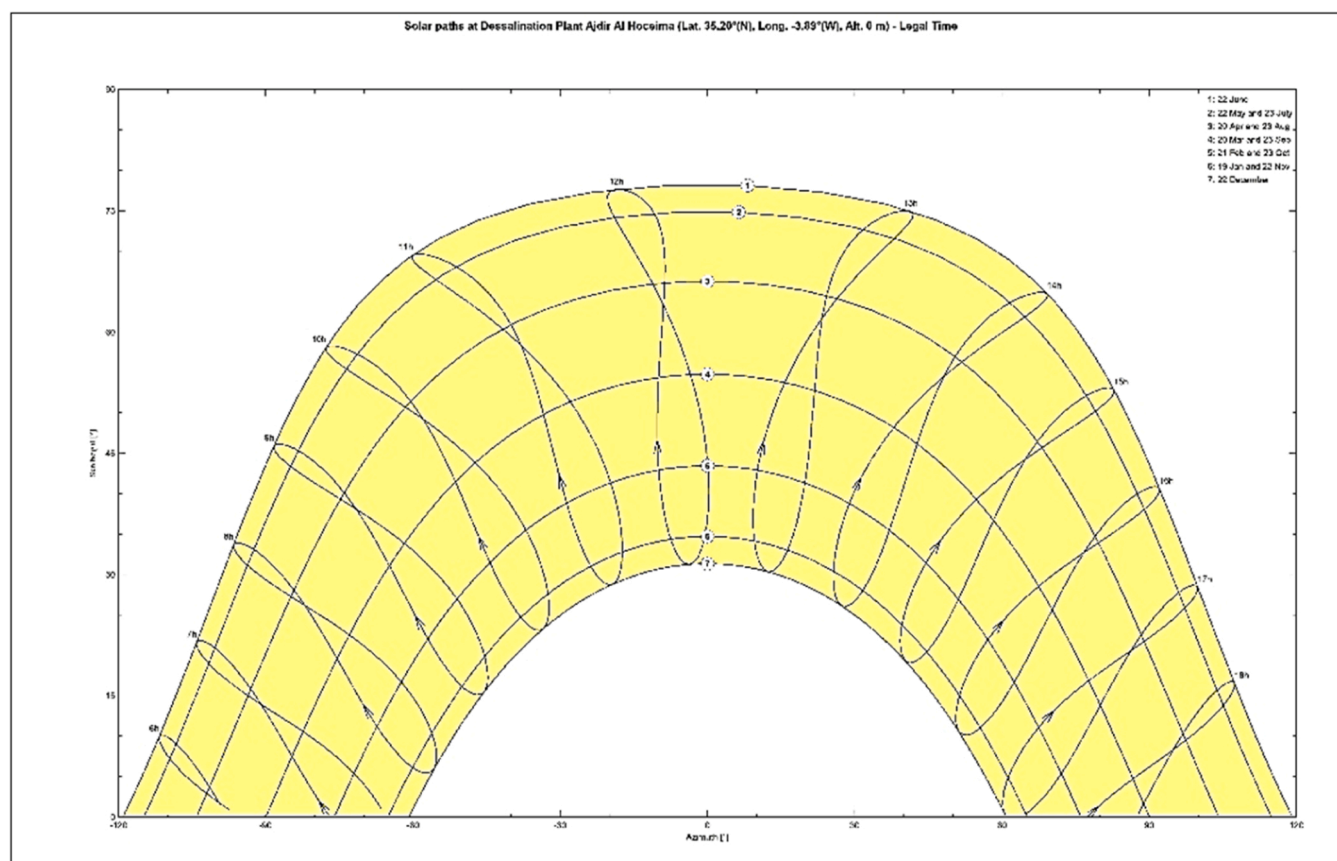


Fig. 10. Sun Path at Ajdir (location of desalination station).

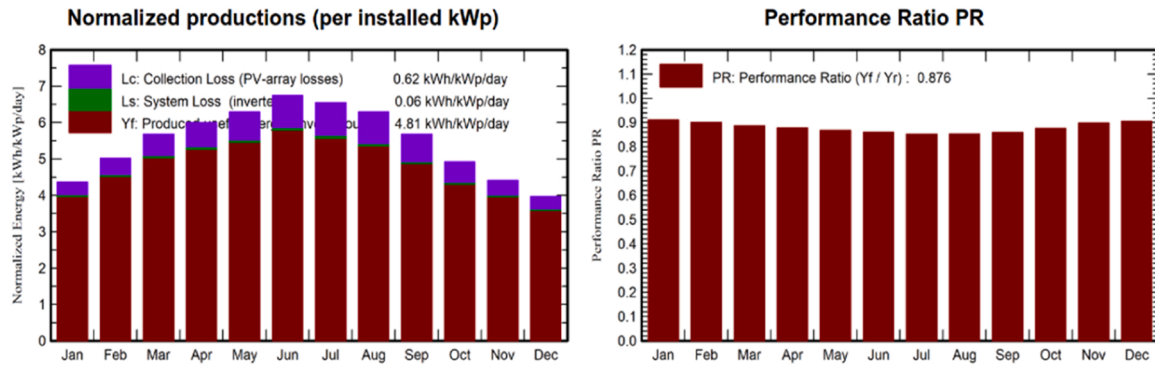


Fig. 11. Annual normalized production and performance ratio of the system.

$$\text{Specific Production} = \frac{\text{Annual Energy Production}}{\text{Peak Power}} \times 1000$$

These results are promising for the integration of solar energy into the desalination plant, as the system is projected to supply a substantial portion of the plant's energy demands, particularly during periods of peak solar irradiance. However, it is essential to acknowledge that these values are derived from theoretical simulations, and actual performance may differ under real-world conditions. To obtain a more accurate assessment of the system's operational efficiency and its impact on reducing the plant's reliance on conventional energy sources, further validation through field testing and real-time monitoring is recommended. Such validation would offer a more precise understanding of the system's performance in practice and its potential for long-term integration.

Regarding the daily input/output diagram, it is observed that the data points are concentrated within the range of 6000 kWh/day to 8500 kWh/day (Fig. 12). This indicates that the electrical energy fed into the grid exceeds 6000 kWh/day for the majority of the year.

4.2.6. Loss diagram

While solar energy offers a promising renewable energy source, its conversion into electricity is not entirely efficient. Various factors contribute to energy losses, reducing the overall output of photovoltaic

systems. This diagram illustrates the various factors that impact the electricity generation of our solar system (Fig. 13). It is clear that these losses are numerous and can be quite substantial. They stem from technical aspects, such as component efficiency and temperature, as well as environmental conditions, including solar irradiance and shading.

The photovoltaic system's energy losses were systematically categorized into irradiation, conversion, and inverter inefficiencies, with quantified impacts on overall efficiency. Initial global horizontal irradiation (1768 kWh/m²) was enhanced by 13.5 % (1967 kWh/m²) through optimal collector plane orientation, partially offset by a 1.9 % reduction from the Incidence Angle Modifier (IAM) factor, which accounts for angular irradiance deviations. Subsequent PV conversion losses under Standard Test Conditions included reductions due to sub-optimal irradiance levels (−0.4 %), temperature-induced efficiency degradation (−7.4 %), electrical mismatch (−2.1 %), and ohmic wiring dissipation (−1.1 %), cumulatively lowering the array's nominal energy output (2540.5 MWh) by 10 % to 2285.5 MWh. Additionally, Inverter losses further during operation (−1.3 %), culminating in a final grid-injected energy of 2256 MWh. Overall, the system retained 88 % of its STC-rated energy potential, with temperature and irradiance variability representing dominant loss mechanisms, while inverter inefficiencies under non-ideal operating conditions underscored the need for advanced maximum power point tracking (MPPT) and thermal

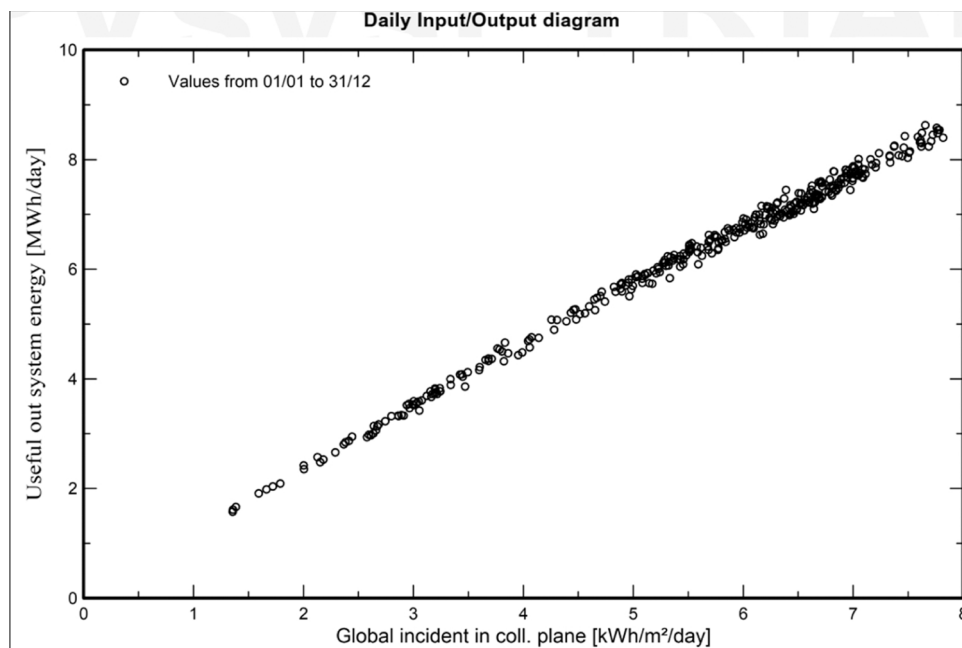


Fig. 12. Diagram of Daily Energy Input to the Grid.

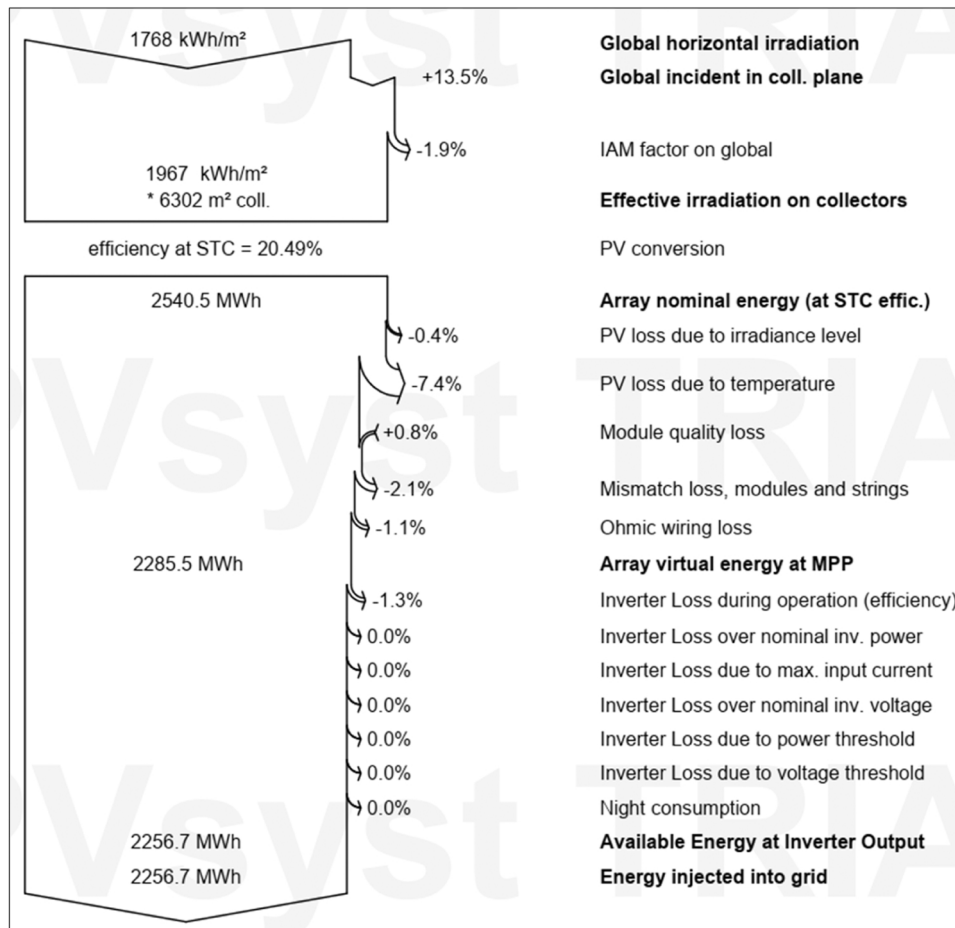


Fig. 13. Simulation of Loss Diagram.

management strategies to optimize yield.

4.2.7. Economic study

The installation of a photovoltaic system comprises two distinct phases: the installation itself and the connection to the electrical network. The latter encompasses wiring, support modules, and other related tasks. The Fig. 14 illustrate the total cost of our system.

- **Calculation of Annual Project Gain**

Based on the simulation results, the annual energy production of the installation is estimated to be 2,256,000 kWh/year. This figure can be multiplied by the unit price of the kWh (calculated as 0.9679 DH/kWh according to ONEE) to obtain the following estimate:

Annual gain = Energy produced × Unit price

Annual gain = 2256000 × 0,9679 = 2183582,4DH/Year

- **Rotor on Investment (ROI)**

The return on investment time is defined as the quotient of the sum of investments to the annual project gain. The total cost of the project is equivalent to the sum of the installation cost and the operating cost (1euro= 10,50 DH). Although the total initial investment cost is in the range of 11874471 DH, this amount includes 11,837,721 DH for installation costs and 36,750 DH for operating costs. The return on investment time becomes:

$$ROI = \frac{\text{Initial Investment}}{\text{Annual gain}}$$

$$ROI = \frac{11874471}{2183582,4} = 5,44\text{years}$$

Economically, the system exhibits notable viability, with a return on

investment (ROI) of 5.44 years. Post-payback, the system is projected to generate net revenue for nearly 25 years, under the assumption of stable energy prices, with a competitive leveled cost of energy (LCOE) of 0.027 EUR/kWh—a figure that is notably lower than Morocco's average residential grid tariff. This cost advantage positions solar energy as a financially sustainable alternative to fossil fuels. Sensitivity analyses indicate that rising electricity prices or policy incentives, such as carbon credits or tax reductions, could further enhance the return on investment (ROI). However, it is imperative to note the inherent challenges posed by the gradual degradation of solar panels, which is estimated to be approximately 0.5 % per annum. This necessitates the initial oversizing of the system to ensure sufficient output over the long term. The macroeconomic benefits of this transition include a reduction in the reliance on imported fossil fuels, an enhancement of Morocco's energy security, and the creation of employment opportunities in the installation, maintenance, and local manufacturing sectors.

4.2.8. Environmental study

Electricity generation through photovoltaic technology results in the emission of pollutants, which can subsequently impact the environment. However, the influence of these gaseous emissions is generally considered minimal during the energy transformation process. Additionally, the installation of photovoltaic panels for electricity production is deemed to be entirely silent and does not generate any noise pollution. It is estimated that the production of each kilowatt-hour (kWh) of electricity results in the emission of 0728 kg of CO₂. Consequently, upon implementing this project, which is expected to generate approximately 2,256,000 kWh annually, it is possible to avoid a CO₂ emission of:

$$CO_2\text{reduction} = \text{Energy Produced} \times \text{kg equivalentCO}_2$$

Cost of the system			
Installation costs			
Item	Quantity units	Cost EUR	Total EUR
PV modules			
JKM-410M-72H-V	3132	180.00	563 760.00
Supports for modules	3132	90.00	281 880.00
Inverters			
Sunny Highpower SHP150-US-21-PEAK3	8	6 799.00	54 392.00
Other components			
Accessories, fasteners	3000	20.00	60 000.00
Wiring	1000	3.50	3 500.00
Combiner box	8	100.00	800.00
Monitoring system, display screen	1	250.00	250.00
Surge arrester	7	90.00	630.00
Studies and analysis			
Engineering	1	2 000.00	2 000.00
Installation			
Global installation cost per module	3132	50.00	156 600.00
Global installation cost per inverter	8	80.00	640.00
Transport	4	200.00	800.00
Grid connection	1	150.00	150.00
Land costs			
Land preparation	1	2 000.00	2 000.00
		Total	1 127 402.00
		Depreciable asset	960 032.00

Operating costs	
Item	Total EUR/year
Maintenance	
Repairs	1 000.00
Cleaning	2 500.00
Total (OPEX)	3 500.00

System summary	
Total installation cost	1 127 402.00 EUR
Operating costs	3 500.00 EUR/year
Produced Energy	2257 MWh/year
Cost of produced energy (LCOE)	0.0270 EUR/kWh

Fig. 14. Total Cost of the System.

$$\text{CO}_2\text{reduction} = 2256000 \times 0.728 = 1642,37\text{ton/year}$$

The Fig. 15 illustrates the variation in CO₂ emissions calculated using PVsyst. From an environmental perspective, the photovoltaic system offers substantial advantages, notably in terms of its substantial reduction in greenhouse gas emissions. It prevents the emission of approximately 1642,37 tons of CO₂ per year, which is comparable to removing around 350 gasoline-powered vehicles from the road annually. Over the span of a 30-year operational lifespan, this translates into a cumulative avoidance of approximately 49271 tons of CO₂ emissions. This underscores the pivotal role of the photovoltaic system in climate change mitigation and its alignment with both Morocco's renewable energy strategy and broader global climate commitments. Moreover, the generation of solar electricity circumvents the emission of harmful

pollutants such as sulfur oxides (SO_x) and nitrogen oxides (NO_x), consequently diminishing the risks of acid rain and respiratory ailments. The system's silent operation facilitates its integration into urban and residential environments, and its installation on rooftops or non-arable land minimizes ecological disruption relative to conventional energy infrastructures. Despite the energy-intensive manufacturing of photovoltaic components, the system demonstrates a short carbon payback period, typically ranging from one to three years, after which it delivers net environmental benefits. Future advancements in panel recycling may further decrease its ecological footprint, reinforcing its contribution to a circular and sustainable energy economy.

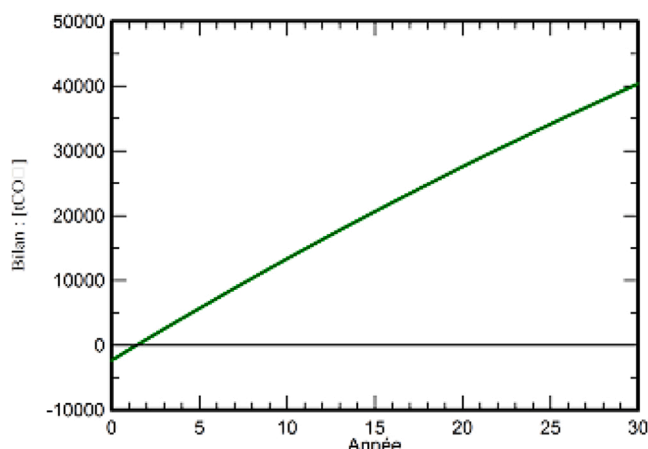


Fig. 15. CO₂ saved according to time.

5. Conclusion

As a conclusion, in this article, we present the results of a study aimed at utilizing photovoltaic solar energy to cover a portion of the electricity consumption of the Al-Hoceima seawater desalination plant. We focused on the reverse osmosis unit as it represents the most energy-intensive part of the plant, consuming over 53 % of the total energy used by the facility. Based on this consumption and the average global solar radiation in Ajdir, we determined the total peak power to be installed, which is 1284 kWp. Using the PVSyst software, we calculated the total surface area required based on the characteristics of the available PV modules. After sizing the solar system, a simulation of the annual electricity production and system losses was performed. This includes presenting the system losses and the amount of CO₂ saved throughout the installation's lifespan. Finally, both an economic and an environmental analysis are presented. The economic study shows that the project will become profitable after 5.58 years, while the environmental analysis outlines the emissions generated and avoided by the installed system. The implementation of this solar system with optimized dimensions has led to a notable reduction in the total energy consumption of the desalination plant at Al-Hoceima, amounting to approximately 10 % of the plant's overall energy usage.

CRediT authorship contribution statement

Salima Boudrahman: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Ibrahim Alsayer:** Software, Methodology, Formal analysis. **Mabrouki Jamal:** Writing – review & editing, Visualization, Validation, Supervision. **Ait Oujallal Ilham:** Writing – original draft, Validation, Methodology, Data curation. **Younes Abrouki:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation.

Declaration of Competing Interest

All of the authors were involved in the design, or analysis and interpretation of the data; the authors wrote the article or revised it critically for important intellectual content; and approval of the final version. This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript

Acknowledgement

The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this

research work through the project number “NBU-FFR-2025-2148-05”.

Data availability

No data was used for the research described in the article.

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